

Shear modulus

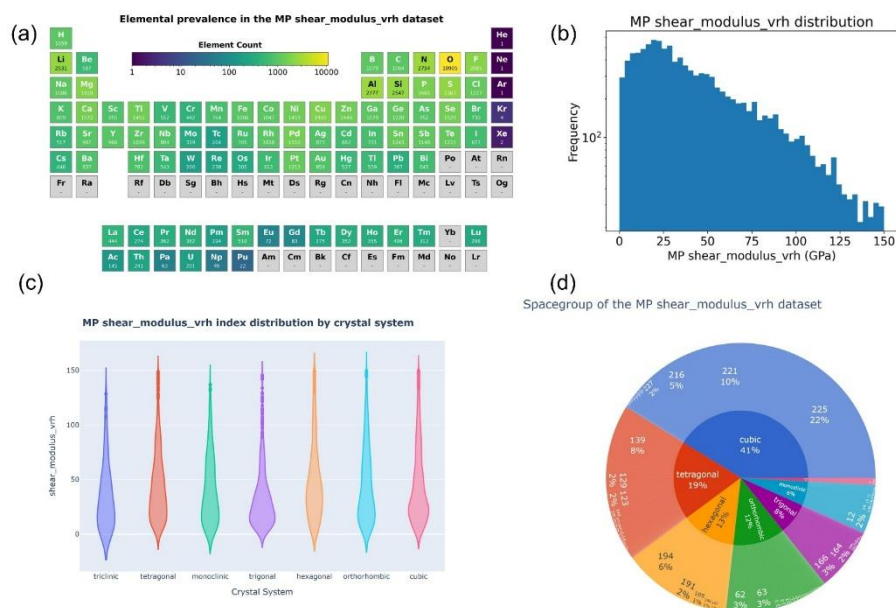


Figure S1 presents the visualization analysis results of 10,812 shear modulus structures selected from the MP database as a general test dataset. These structures were extracted from 12,334 raw structures and include: (a) Visualization of element distribution; (b) Histogram of shear modulus values; (c) Distribution of shear modulus values based on crystal system classification; (d) Pie chart of shear modulus dataset classified by crystal system and corresponding space groups.

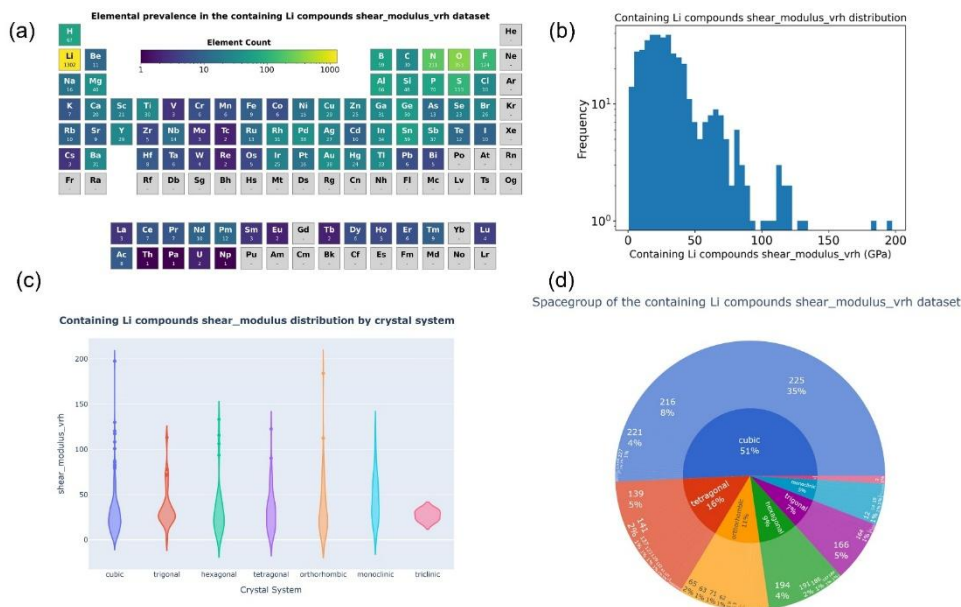


Figure S2 presents the visualization analysis results of 422 structures extracted from the remaining lithium-containing structures in the MP shear modulus dataset after excluding the general test dataset, serving as a lithium compound test dataset. The contents include: (a) Visualization of element distribution; (b) Histogram of shear modulus values; (c) Distribution of shear modulus values based on crystal system classification; (d) Pie chart of shear modulus dataset classified by crystal system and corresponding space groups.

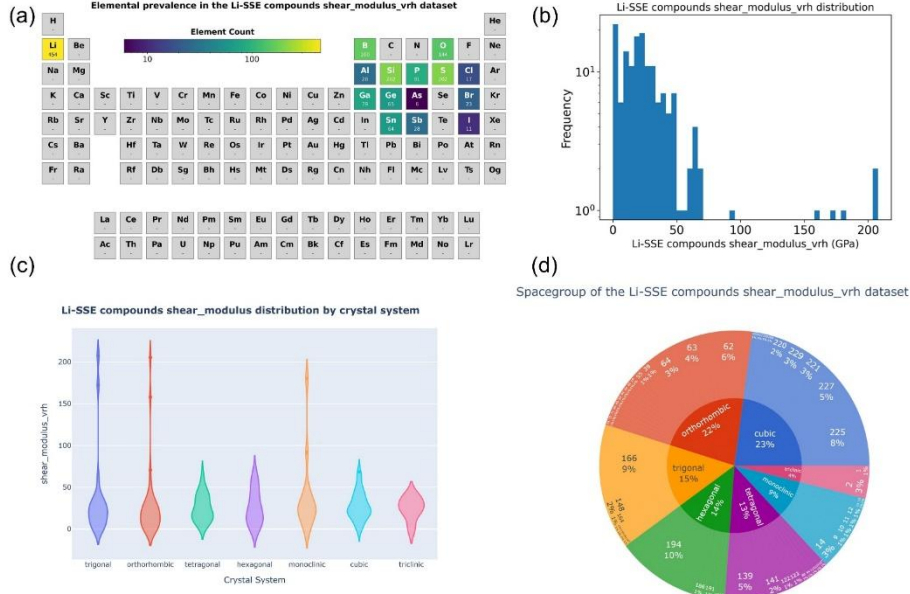


Figure S3 presents the visualization analysis results of 153 structures extracted from the remaining lithium-containing structures in the MP shear modulus dataset after excluding the general test dataset, serving as a lithium solid electrolyte test dataset. The contents include: (a) Visualization of element distribution; (b) Histogram of shear modulus values; (c) Distribution of shear modulus values based on crystal system classification; (d) Pie chart of shear modulus dataset classified by crystal system and corresponding space groups.

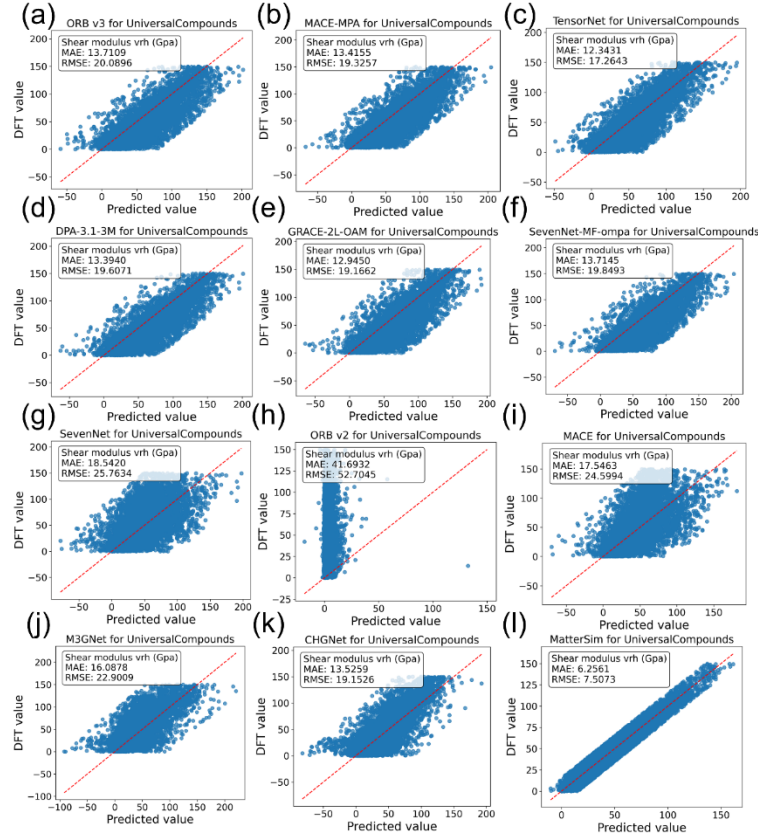


Figure S4 evaluates the performance of 12 MLIPs on the MP shear modulus universal test dataset, comparing their predictions with DFT results.

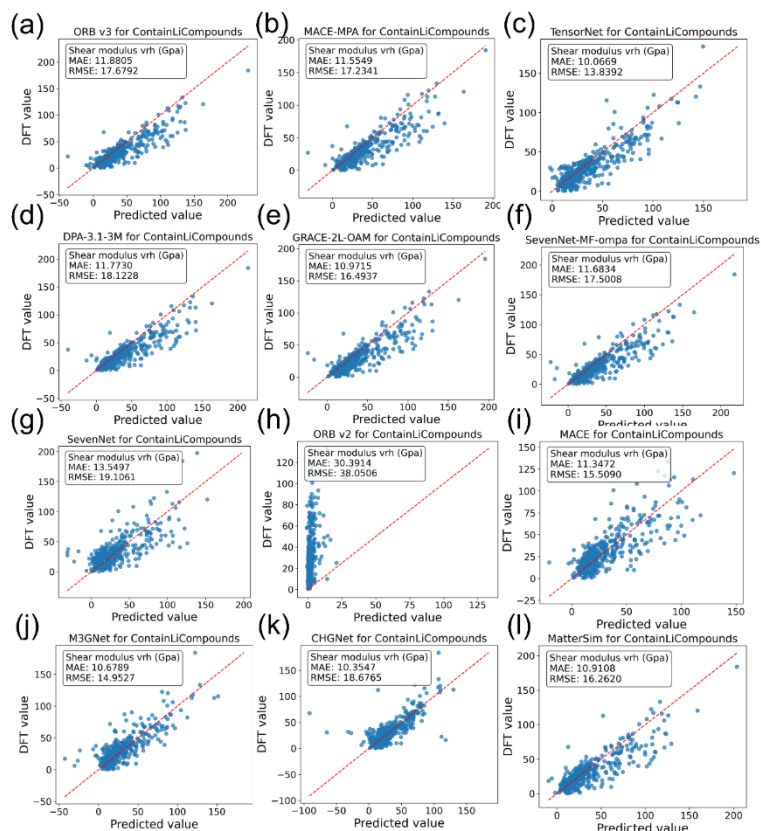


Figure S5 evaluates the performance of 12 MLIPs on the MP shear modulus lithium compound test dataset, comparing their predictions with DFT results. .

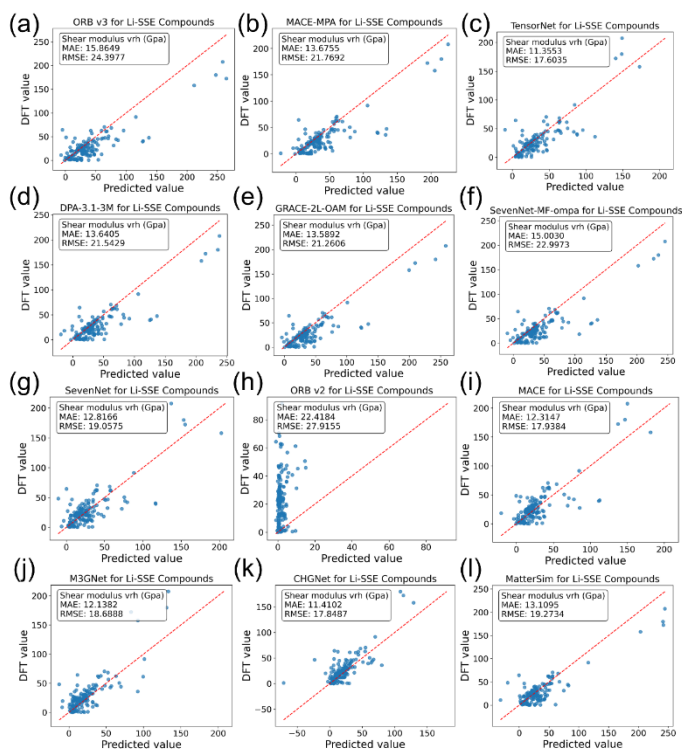


Figure S6 presents the performance of 12 MLIPs on the MP shear modulus lithium solid electrolyte test dataset, comparing their predictions with DFT results.

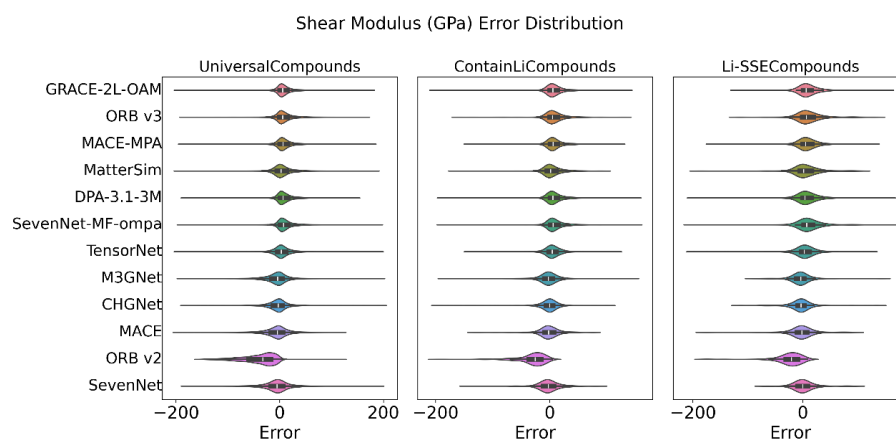


Figure S7 presents error distribution plots comparing the predictions of 12 MLIPs with DFT results on the UniversalCompounds, ContainLiCompounds, and Li-SSECompounds datasets (three levels of the MP shear modulus test set).